

Homework Week 4

QMIR 2026 – Homework 04

Charel Frisch

March 12, 2026

Table of contents

Setup	2
Exercise 1 – Import and Explore	2
Inspect the Data	2
Subset to Parliament Elections	5
Exercise 2 – Filter and Select	6
German Elections	6
Recent Western Europe	7
Exercise 3 – Mutate and Compute New Variables	8
Seat Share	8
Vote–Seat Disproportionality	9
Exercise 4 – Summarise and Group	10
Mean Vote Share by Party Family	10
Electoral Volatility Over Time	12
Exercise 5 – A Full Pipeline	13
Exercise 6 – A Labelled Figure	15
Exercise 7 – Reflection	18

Setup

Load the packages you need and import the data. The ParlGov dataset is an Excel file with multiple sheets. Use `read_excel()` from the `readxl` package and specify the sheet with the `sheet` argument.

```
library(tidyverse)
library(readxl)

# Import the election sheet
parlgov_election <- read_excel("parlgov.xlsx", sheet = "election")

# Import the party sheet -- you will need it later
parlgov_party <- read_excel("parlgov.xlsx", sheet = "party")
```

Add `library()` calls for any additional packages you use.

Exercise 1 – Import and Explore

Inspect the Data

Use `glimpse()`, `head()`, and `summary()` to get an overview of `parlgov_election`. Then answer the questions below in prose.

```
# Your exploration code here

glimpse(parlgov_election)
```

```
Rows: 9,016
Columns: 16
$ country_name_short    <chr> "AUS", "AUS", "AUS", "AUS", "AUS", "AU~
$ country_name          <chr> "Australia", "Australia", "Australia",~
$ election_type         <chr> "parliament", "parliament", "parliamen~
$ election_date         <chr> "1901-03-30", "1901-03-
30", "1901-03-3~
$ vote_share            <dbl> 44.4, 34.2, 19.4, 1.4, 0.6, 29.7, 34.4~
$ seats                 <dbl> 32, 26, 15, 1, 1, 26, 25, 23, 1, 0, 27~
$ seats_total           <dbl> 75, 75, 75, 75, 75, 75, 75, 75, 75, 75~
```

```

$ party_name_short      <chr> "PP", "FTP", "ALP", "none", "one-
seat"~
$ party_name            <chr> "Protectionist Party", "Free Trade Par~
$ party_name_english    <chr> "Protectionist Party", "Free Trade Par~
$ left_right            <dbl> 7.4000, 6.0000, 3.8833, NA, NA, 7.4000~
$ country_id            <dbl> 33, 33, 33, 33, 33, 33, 33, 33, 33, 33~
$ election_id           <dbl> 731, 731, 731, 731, 731, 730, 730, 730~
$ previous_parliament_election_id <dbl> NA, NA, NA, NA, NA, 731, 731, 731, 731~
$ previous_cabinet_id   <dbl> NA, NA, NA, NA, NA, 997, 997, 997, 997~
$ party_id              <dbl> 1898, 1938, 1253, 1396, 2299, 1898, 19~

```

```
head(parlgov_election)
```

```

# A tibble: 6 x 16
  country_name_short country_name election_type election_date vote_share seats
  <chr>              <chr>      <chr>      <chr>          <dbl> <dbl>
1 AUS                Australia parliament  1901-03-30      44.4    32
2 AUS                Australia parliament  1901-03-30      34.2    26
3 AUS                Australia parliament  1901-03-30      19.4    15
4 AUS                Australia parliament  1901-03-30       1.4     1
5 AUS                Australia parliament  1901-03-30       0.6     1
6 AUS                Australia parliament  1903-12-16      29.7    26
# i 10 more variables: seats_total <dbl>, party_name_short <chr>,
#   party_name <chr>, party_name_english <chr>, left_right <dbl>,
#   country_id <dbl>, election_id <dbl>, previous_parliament_election_id <dbl>,
#   previous_cabinet_id <dbl>, party_id <dbl>

```

```
summary(parlgov_election)
```

country_name_short	country_name	election_type	election_date
Length:9016	Length:9016	Length:9016	Length:9016
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

vote_share	seats	seats_total	party_name_short
Min. : 0.00	Min. : 0.00	Min. : 5.0	Length:9016
1st Qu.: 2.10	1st Qu.: 1.00	1st Qu.: 81.0	Class :character
Median : 6.50	Median : 6.00	Median :152.0	Mode :character

```

Mean      :11.79   Mean      : 23.21   Mean      :213.5
3rd Qu.:17.50   3rd Qu.: 24.00   3rd Qu.:265.0
Max.      :71.20   Max.      :470.00   Max.      :736.0
NA's      :560     NA's      :64

  party_name      party_name_english  left_right      country_id
Length:9016      Length:9016        Min.      :0.000   Min.      : 1.0
Class :character  Class :character   1st Qu.:3.300   1st Qu.:21.0
Mode  :character  Mode  :character   Median :5.714   Median :37.0
                                   Mean  :5.146   Mean  :37.2
                                   3rd Qu.:7.240   3rd Qu.:55.0
                                   Max.  :9.825   Max.  :75.0
                                   NA's  :947

  election_id      previous_parliament_election_id previous_cabinet_id
Min.      : 1.0    Min.      : 1.0                      Min.      : 2.0
1st Qu.: 306.0    1st Qu.: 293.0                      1st Qu.: 361.0
Median : 604.0    Median : 581.0                      Median : 752.0
Mean      : 590.1   Mean      : 562.8                      Mean      : 786.3
3rd Qu.: 866.0    3rd Qu.: 804.0                      3rd Qu.:1156.0
Max.      :1124.0  Max.      :1122.0                      Max.      :1696.0
NA's      :297     NA's      :283

  party_id
Min.      : 2
1st Qu.: 535
Median :1030
Mean      :1142
3rd Qu.:1577
Max.      :2909

```

```
parlgov_election |> distinct(country_name)
```

```

# A tibble: 37 x 1
  country_name
  <chr>
1 Australia
2 Austria
3 Belgium
4 Bulgaria
5 Canada
6 Switzerland
7 Cyprus
8 Czech Republic

```

```
9 Germany
10 Denmark
# i 27 more rows
```

```
parlgov_election |> count(election_type)
```

```
# A tibble: 2 x 2
  election_type      n
  <chr>         <int>
1 ep           1702
2 parliament    7314
```

Questions – answer in prose beneath this line:

- How many rows and columns does the election sheet have? What does one row represent?
- What data types are the columns? Are there any that surprise you?
- How many distinct countries are in the dataset? Use `distinct()`.
- Use `count()` to find out how many observations belong to parliamentary elections vs. European Parliament elections. Which type is more common?

The election sheet has 9,016 rows and 16 columns where as one column represent one variable and one row represents basically one party and its respective results and information. There are 37 distinct countries in the dataset, 1702 observations belong to the EP-election and 7314 observations belong to Parliament elections. Therefore, parliamentary elections are more common

Subset to Parliament Elections

For the rest of the homework (unless noted otherwise), work only with **national parliament elections**. Filter `parlgov_election` to keep only rows where `election_type == "parliament"` and assign the result to a new object called `parl`.

```
parl <- parlgov_election |>
  filter(election_type == "parliament")
```

How many rows remain?

```
# Your check here

glimpse(parl)
```

```

Rows: 7,314
Columns: 16
$ country_name_short      <chr> "AUS", "AUS", "AUS", "AUS", "AUS", "AU~
$ country_name            <chr> "Australia", "Australia", "Australia",~
$ election_type           <chr> "parliament", "parliament", "parliamen~
$ election_date           <chr> "1901-03-30", "1901-03-
30", "1901-03-3~
$ vote_share              <dbl> 44.4, 34.2, 19.4, 1.4, 0.6, 29.7, 34.4~
$ seats                   <dbl> 32, 26, 15, 1, 1, 26, 25, 23, 1, 0, 27~
$ seats_total             <dbl> 75, 75, 75, 75, 75, 75, 75, 75, 75, 75~
$ party_name_short        <chr> "PP", "FTP", "ALP", "none", "one-
seat"~
$ party_name              <chr> "Protectionist Party", "Free Trade Par~
$ party_name_english      <chr> "Protectionist Party", "Free Trade Par~
$ left_right              <dbl> 7.4000, 6.0000, 3.8833, NA, NA, 7.4000~
$ country_id              <dbl> 33, 33, 33, 33, 33, 33, 33, 33, 33, 33~
$ election_id             <dbl> 731, 731, 731, 731, 731, 730, 730, 730~
$ previous_parliament_election_id <dbl> NA, NA, NA, NA, NA, 731, 731, 731, 731~
$ previous_cabinet_id     <dbl> NA, NA, NA, NA, NA, 997, 997, 997, 997~
$ party_id                <dbl> 1898, 1938, 1253, 1396, 2299, 1898, 19~

```

As expected, only 7314 rows remain. (I think glimpse() was a bit superfluous here because I could've just looked in the running session under DATA)

Exercise 2 – Filter and Select

German Elections

Create an object called `germany` containing all parliamentary election results for Germany. Use `distinct()` on `election_date` to find out how many elections are covered.

```

germany <- parl |>
  filter(country_name == "Germany")

# How many elections?

germany |>
  distinct(election_date)

```

```
# A tibble: 29 x 1
  election_date
  <chr>
1 1919-01-19
2 1920-06-06
3 1924-05-04
4 1924-12-07
5 1928-05-20
6 1930-09-14
7 1932-07-31
8 1932-11-06
9 1933-03-05
10 1949-08-14
# i 19 more rows
```

There are 29 elections covered in the data set

Recent Western Europe

From `parl`, filter to keep only observations from **France, Germany, the Netherlands, and Sweden** from the year **2000 onwards**.

You will need to extract the year from `election_date`. Because the date is stored as a character string (or date object), you can use `year(as.Date(election_date))` inside `mutate()` to create a numeric year column first – or use `stringr::str_sub(election_date, 1, 4)` to extract the first four characters as a string and wrap it in `as.numeric()`.

Then keep only the columns `country_name`, `election_date`, `party_name_english`, `vote_share`, `seats`, `seats_total`, and `left_right`.

```
recent_we <- parl |>
  mutate(year = as.numeric(stringr::str_sub(election_date, 1, 4))) |> # extract year
  filter(
    country_name %in% c("France", "Germany", "Netherlands", "Sweden"),
    year >= 2000
  ) |>
  select(country_name,
election_date,
party_name_english,
vote_share,
seats,
seats_total,
```

```
left_right)
```

```
recent_we
```

```
# A tibble: 269 x 7
```

	country_name	election_date	party_name_english	vote_share	seats	seats_total
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	Germany	2002-09-22	Social Democratic Pa~	38.5	251	603
2	Germany	2002-09-22	Christian Democratic~	29.5	190	603
3	Germany	2002-09-22	Christian Social Uni~	9	58	603
4	Germany	2002-09-22	Alliance 90 / Greens	8.6	55	603
5	Germany	2002-09-22	Free Democratic Party	7.4	47	603
6	Germany	2002-09-22	PDS The Left	4	2	603
7	Germany	2002-09-22	The Republicans	0.6	0	603
8	Germany	2005-09-18	Social Democratic Pa~	34.2	222	614
9	Germany	2005-09-18	Christian Democratic~	27.8	180	614
10	Germany	2005-09-18	Free Democratic Party	9.8	61	614

```
# i 259 more rows
```

```
# i 1 more variable: left_right <dbl>
```

Write one sentence confirming the result looks as expected.

The new set contains 269 rows from elections since the years 2000. Every row represents a party election result, so 269 rows seems to be about correct considering it is 4 countries in total.

Exercise 3 – Mutate and Compute New Variables

Seat Share

In `parl`, create a new variable `seat_share` that expresses the share of seats won by each party as a percentage (i.e., `seats / seats_total * 100`). Assign the result back to `parl`.

```
parl <- parl |>
  mutate(seat_share = seats / seats_total * 100)
```

Now use `filter()` and `arrange()` to find the five parties with the highest single-election seat share in the entire dataset. What do they have in common?


```
parl |>
  filter(!is.na(seat_share)) |>
  arrange(desc(seat_share)) |>
  select(country_name, election_date, party_name_english, seat_share) |>
  slice_head(n = 5)
```

```
# A tibble: 5 x 4
```

	country_name <chr>	election_date <chr>	party_name_english <chr>	seat_share <dbl>
1	Canada	1958-03-31	Progressive Conservative Party of Can~	78.5
2	United Kingdom	1931-10-27	Conservatives	76.4
3	Canada	1984-09-04	Progressive Conservative Party of Can~	74.8
4	Canada	1940-03-26	Liberal Party of Canada	73.9
5	Greece	1974-11-17	New Democracy	73.3

I am not sure if I am supposed to use a command here but I know that all of these countries use some form of the first-past-the-post election system (Westminster, etc.)

Vote–Seat Disproportionality

A classic question in electoral systems research is whether parties receive seat shares that are proportional to their vote shares, or whether the electoral system distorts the translation of votes into seats.

Create a variable `disproportionality` defined as `seat_share - vote_share`. Positive values mean a party received more seats than its vote share would proportionally warrant; negative values mean fewer.

Then compute the mean disproportionality **by country**, keeping only countries with at least 50 observations (use `filter(n() >= 50)` inside a `group_by()` chain). Sort by mean disproportionality descending.

```
parl |>
  mutate(disproportionality = seat_share - vote_share) |>
  filter(!is.na(disproportionality)) |>
  group_by(country_name) |>
  filter(n() >= 50) |>
  summarise(
    mean_disprop = round(mean(disproportionality), 2),
    n = n(),
```

```

    .groups = "drop"
  ) |>
  arrange(desc(mean_disprop))

```

```

# A tibble: 37 x 3
  country_name mean_disprop     n
  <chr>         <dbl> <int>
1 Bulgaria         0.48   132
2 New Zealand      0.44   189
3 Portugal         0.39   100
4 Greece           0.35   138
5 Latvia           0.29   107
6 Turkey           0.29    76
7 Estonia          0.28    79
8 Spain            0.28   197
9 Belgium          0.26   322
10 Slovakia        0.21   107
# i 27 more rows

```

Which countries show the largest average disproportionality? Can you explain why based on what you know about their electoral systems? Write two to three sentences.

Funnily enough, this was the hardest part of the exercise. The countries with the largest disproportionalities are Bulgaria, New Zealand, Portugal, Greece and Latvia. NZ, Portugal and Greece are quite easy to explain because until the 1990s NZ used the Westminster system and Greece has some form of FPTP until today. Sadly, I am not that much into Bulgarian and Latvian Politics so ... I looked it up and apparently Bulgaria used FPTP for 2 elections but I couldn't find anything so I gave up after 20 minutes. If you know why, please let me know because now I am really interested! :(

Exercise 4 – Summarise and Group

Mean Vote Share by Party Family

The `party` sheet contains a `family_name` variable that classifies parties into ideological families (e.g., Social democracy, Conservative, Green/Ecologist).

The `parl` (election) object also contains a `left_right` variable at the party-election level. Use it to compute the mean and standard deviation of `left_right` by country,

for national parliament elections only. Remove missing values with `na.rm = TRUE`. Sort by mean left-right score ascending (most left-leaning countries first) and keep only countries with at least 30 non-missing observations.

```
parl |>
  filter(!is.na(left_right)) |>
  group_by(country_name) |>
  summarise(
    mean_lr = round(mean(left_right, na.rm = TRUE), 2),
    sd_lr = round(sd(left_right, na.rm = TRUE), 2),
    n = n(),
    .groups = "drop"
  ) |>
  filter(n >= 30) |>
  arrange(mean_lr)
```

```
# A tibble: 37 x 4
  country_name mean_lr sd_lr      n
  <chr>         <dbl> <dbl> <int>
1 Portugal      3.88  2.41  142
2 Spain         4.58  2.26  176
3 Greece        4.63  2.64  138
4 Iceland       4.68  2.08  180
5 Slovenia      4.72  1.98  107
6 Ireland       4.74  2.07  174
7 Norway        4.78  2.39  238
8 Switzerland   4.85  2.62  344
9 Luxembourg    4.91  2.49  145
10 Japan        4.93  2.59  186
# i 27 more rows
```

Write one sentence interpreting the variation in mean left-right scores across countries. Is the pattern what you expected?

The pattern isn't at all what I expected except for maybe Spain, Portugal and Norway which are well-known to be left-leaning... but Japan and Luxembourg??? Two countries that have in common that they have been ruled for 70 years straight by conservative parties, that was really surprising.

Electoral Volatility Over Time

Compute the mean vote share of the **largest party** (maximum `vote_share`) per election, grouped by decade. To do this:

1. Create a `decade` variable using `mutate()`: `(year %/% 10) * 10` where `year` is the numeric year you extracted earlier (or extract it again here).
2. For each election (identified by `election_date` and `country_name`), find the maximum vote share.
3. Then group by decade and compute the mean of those election-level maxima.

This tells you whether dominant parties have become stronger or weaker over time.

```
parl |>
  mutate(
    year = as.numeric(str_sub(election_date, 1, 4)),
    decade = (year %/% 10) * 10
  ) |>
  group_by(country_name, election_date, decade) |>
  summarise(max_vote = max(vote_share, na.rm = TRUE), .groups = "drop") |>
  group_by(decade) |>
  summarise(
    mean_max_vote = round(mean(max_vote, na.rm = TRUE), 1),
    n_elections = n(),
    .groups = "drop"
  ) |>
  filter(n_elections >= 5) |>
  arrange(decade)
```

A tibble: 13 x 3

	decade	mean_max_vote	n_elections
	<dbl>	<dbl>	<int>
1	1900	45.4	27
2	1910	-Inf	40
3	1920	36.9	57
4	1930	39.4	48
5	1940	40.5	49
6	1950	40.7	66
7	1960	41.1	59
8	1970	38.6	73
9	1980	-Inf	76
10	1990	-Inf	105
11	2000	34.7	98

12	2010	32.6	110
13	2020	30.6	37

Write two sentences describing the trend. Is there evidence of increasing or decreasing party system fragmentation over time?

If we look at the result, there is definitely evidence for an increase in the fragmentation of the party system which is coherent with our observations of modern parliaments. Until the 1980s, Germany had only 3 parties in Parliament, nowadays there are 5. In the Netherlands, it is even more extreme with up to 17 parties.

Exercise 5 – A Full Pipeline

Write a **single piped chain** (no intermediate objects) that answers the following question:

Which party family won the most total seats across all national parliamentary elections in Western European countries (France, Germany, Italy, Spain, the Netherlands, Belgium, Austria, Sweden, Denmark, Norway, Finland, Portugal) since 1990?

Your pipeline should:

1. Filter to the relevant countries and years
2. Filter out missing seat values
3. Group by `family_name` (use the `left_right`-adjacent `party_name_english` — actually, the `parl` table has no `family_name` column directly; you will need to **join** the party family information from `parlgov_party`)

Wait — this is a good moment to introduce joining. The `parl` table does not have `family_name`; that lives in `parlgov_party`. You can bring it in with a left join:

```
# This join adds family_name to parl from parlgov_party
parl_with_family <- parl |>
  left_join(
    parlgov_party |> select(part_id, family_name),
    by = "party_id"
  )
```

Now use `parl_with_family` to complete the pipeline:

```

parl_with_family |>
  mutate(year = as.numeric(str_sub(election_date, 1, 4))) |>
  filter(
    country_name %in%
      c(
        "France",
        "Germany",
        "Italy",
        "Spain",
        "Netherlands",
        "Belgium",
        "Austria",
        "Sweden",
        "Denmark",
        "Norway",
        "Finland",
        "Portugal"
      ),
    year >= 1990,
    !is.na(seats),
    !is.na(family_name)
  ) |>
  group_by(family_name) |>
  summarise(
    total_seats = sum(seats, na.rm = TRUE),
    n_parties = n_distinct(party_name_english),
    .groups = "drop"
  ) |>
  arrange(desc(total_seats))

```

```

# A tibble: 10 x 3
  family_name      total_seats n_parties
  <chr>           <dbl>     <int>
1 Social democracy 10716      28
2 Conservative    6579      34
3 Christian democracy 4631      27
4 Liberal         4243      46
5 Right-wing      2414      37
6 Communist/Socialist 2094      34
7 Green/Ecologist 1438      22
8 Agrarian        930       4
9 no family       442       5

```

Write two to three sentences interpreting the result. Does the ranking reflect your expectations about Western European politics since 1990?

At first I was bit surprised because conservatie parties won considerably more seats than christian democratic parties. However this should be correct because most countries lost their traditional big christian democratic parties over time. Social democratic parties winning the most seats is very much expected considering their dominace in Nordic and Iberian countries. Additionally most of the social democratic parties survived their christian democratic counterparts

Exercise 6 – A Labelled Figure

Produce a figure that visualises something substantively interesting from the ParlGov data. You have freedom here – choose a question that genuinely interests you.

Requirements:

- The figure must be produced with `ggplot2`
- The code cell must have a `label` starting with `fig-` and a `fig-cap` with a descriptive caption
- You must reference the figure in your prose using `@fig-yourlabel`
- Use `theme_bw()` or another clean theme
- Add informative axis labels with `labs()`

Below are three suggested starting points if you need inspiration. You are not required to use them – they are just prompts.

Option A – Ideological polarisation over time: For a country of your choice, compute the standard deviation of `left_right` across parties per election year and plot it as a line over time. A rising line suggests increasing ideological spread between parties.

Option B – Green party performance: Filter to parties in the Green/Ecologist family. Plot their mean vote share by country (using `geom_col()`) for elections since 2000, sorted from highest to lowest.

Option C – Left-right distribution by party family: Use `geom_boxplot()` or `geom_violin()` to show the distribution of `left_right` scores across party families. This gives a visual summary of how well the family labels capture ideological position.

```

# Your ggplot code here

library(dplyr)
library(ggplot2)
library(stringr)
library(scales)

parl_with_family |>
  mutate(
    year = as.numeric(str_sub(election_date, 1, 4)),
    family_name = factor(
      family_name,
      levels = c(
        "Communist/Socialist",
        "Green/Ecologist",
        "Social democracy",
        "Agrarian",
        "Liberal",
        "Christian democracy",
        "Conservative",
        "Right-wing"
      )
    )
  ) |>
  filter(
    year >= 2000,
    !family_name %in% c("to be coded", "Special issue", "no family")
  ) |>
  group_by(year, family_name) |>
  summarise(vote = mean(vote_share, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = year, y = vote, fill = family_name)) +
  geom_area(position = "fill") +
  scale_y_continuous(labels = percent_format()) +
  labs(
    x = "Year",
    y = "Vote Share (%)",
    fill = "Party Family"
  ) +
  theme_minimal()

# I want to clarify that the code for this plot is NOT from myself, I got from "http

```

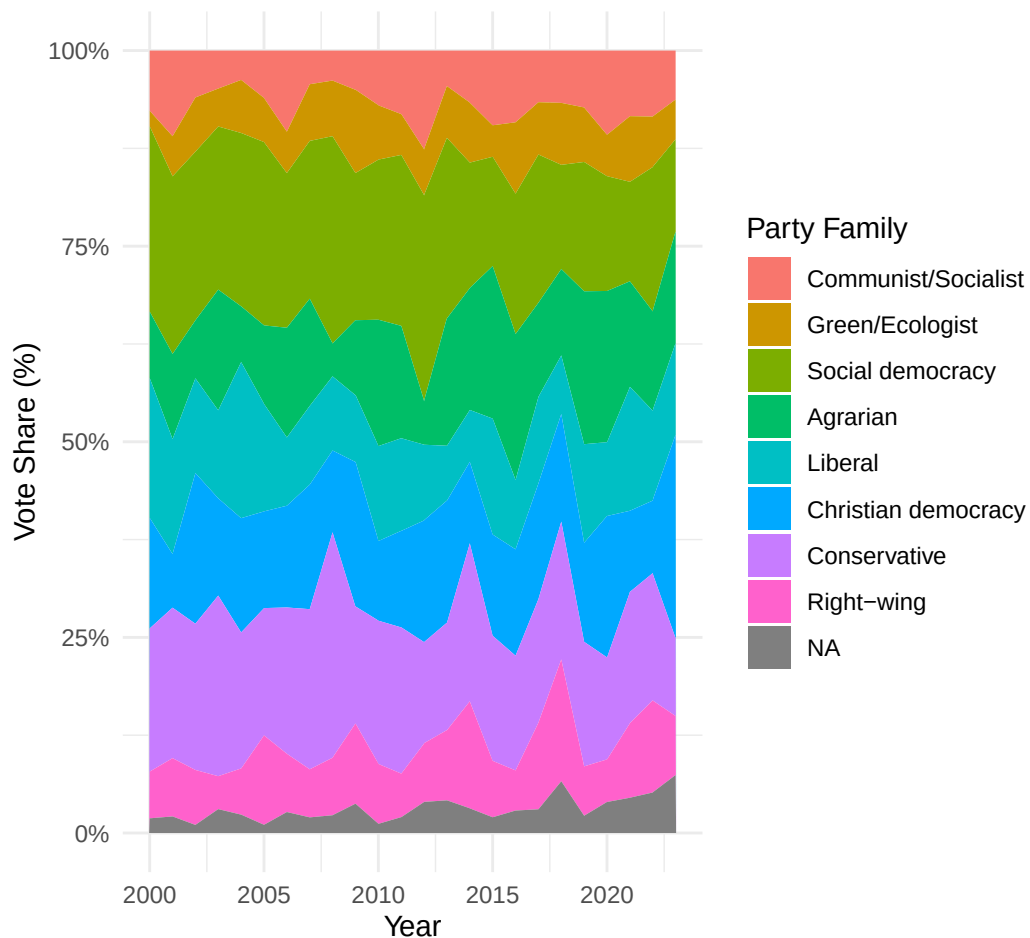



Figure 1: Mean vote share of European party families since the year 2000

Write a paragraph beneath the figure (at least three sentences) interpreting what it shows. Reference the figure using `@fig-yourlabel`. Explain the pattern, note anything surprising, and consider one possible limitation of the visualisation.

Figure 1 shows the political evolution of Europe. We can clearly see that traditional conservative, liberal and social-democratic parties lost quite a big amount of their vote share to smaller party families like right-wing or green parties. Furthermore, one can also see that the vote share of left parties in total massively shrunk from over 35% to under 25%. We can also see that this might correlate with the rise of right-wing parties which previous work in political science has already shown.

Exercise 7 – Reflection

Add a short reflection at the end of your document. Answer the following in three to five sentences:

- Which part of the wrangling workflow was most difficult for you this week?
- Is there anything about the ParlGov data structure that confused you at first?
- What is one thing you now understand about the tidyverse that you did not before this homework?

Be honest – this helps improve the course.

- I think most problems I encountered were due to me doing this at 4am which in hindsight wasn't the brightest idea. Besides that, I actually had a good time doing this assignment. I really like your worksheets, it always feels more like experimenting and playing a bit around than actual work. The plot was obviously the most difficult part even with help from the internet and ChatGPT but I think I figured it out at the end.*